

Pinball Brief 2 – The Silver Ball

In engineering and the technical sciences, the importance of standards cannot be overstated. Organizations ranging from the ASCE in Civil Engineering to IEEE in Computer Science create technical standards that ensure organizational constituency across the world. Standards in technical disciplines may be expected, but they are just as widespread in entertainment-focused industries. Standardized rules in sports, board games and video games allow competitors to translate their skills across different environments. One surprising example of a game where standards have come to govern play is pinball – and specifically, the surprising standards of the silver ball itself.

As with most standards, the unofficial rules governing the size, weight, and composition of the pinball were not always set in stone. The oldest pinball machines from the 1930s did not have any sort of standardization in ball material or size. There were over 100 pinball manufacturers at the time, which only amplified the disorder [12]. Early pinballs were repurposed ball bearings made for appliances like swivel chairs and vacuums. Thus, pinball manufacturers and operators would use ball bearings from wherever it was easiest to obtain in the local area [5]. The comparative lack of a global market contributed to the lack of standardization.

Slowly, out of relative chaos, a standard for pinballs would emerge. Modern pinballs are, with few exceptions, 1 1/16 inches (2.7 cm) and 80 grams (2.82 oz) [3]. Common pinballs are generally grade 25 ball bearings, meaning a tolerance of 0.0005” diametric deviation and 0.00025” sphericity is allowed [15]. The diameter and weight standards have remained unchanged for approximately 60-70 years. Smaller balls are too difficult to see and have the possibility of getting stuck in obstacles, while larger balls are more expensive and cannot fit through the narrow spaces, such as ramps, found on many tables [4]. Thus, 1 1/16th inch pinballs at 80 grams were found to be the ideal structure, and the unofficial-official standard has remained in place. Since most games are designed around this size and weight, pinballs can easily be swapped between games, providing significant cost advantages to players.

Despite the standardization of size and weight, there remains variance in the material composition of the pinball. Most pinballs fall into one of two material categories: carbon steel, or chrome steel [1]. Carbon steel balls are composed of case-hardened, low-carbon steel such as AISI 1010 (~0.1% carbon) or AISI 1020 (~0.2% carbon) [10]. The surface hardness is generally in the range of Rockwell C58-C62, while the internal hardness of the ball is generally lower, between Rockwell C20-C30. Chrome steel balls are composed of high-carbon, chromium-infused, and through-hardened steel such as AISI 52100 (~1% carbon, ~1.5%



Left to right: Carbon steel, Chromium steel, and “Mirror Glazed” pinballs. Notice subtle differences in metallic and reflective properties. [8]

chromium). Manganese and Silicon are also added to improve the depth of hardening. As Chrome steel balls are through-hardened, both the internal and surface hardness range are between Rockwell C60-C66 [8].

The lack of a composition standard stems from the trade-offs between materials. The main advantage to carbon steel balls is their cost. Carbon steel pinballs are the more common of the two categories due to being cheaper to manufacture and purchase [9]. However, they generally have poorer wear resistance and lower hardness compared to chrome steel. Chrome steel pinballs are also easier to polish and to get the “mirror-like” shininess that many pinball players enjoy [14]. Magnetism is also important to consider when using a pinball for a long time. Both chrome and carbon pinballs are ferromagnetic, but carbon steel balls have a lower magnetic permeability. This means that for tables featuring highly magnetic components, such as *Teenage Mutant Ninja Turtles* with its spinning magnetic “pizza” at the center of the playfield, the chrome ball can become undesirably attracted to parts of the table.

A few tables have famously disregarded the usual standards for pinballs and used novel and unexpected ball materials to provide a new experience for players. *Twilight Zone* uses five carbon steel pinballs and a special, nonmagnetic “power ball” that is released during certain parts of the game. Bonus points are awarded when the power ball is in play [6]. The power ball is white, faster, larger, ceramic, and only weighs 65 grams. Since the ceramic ball is not affected by magnets, *Twilight Zone* is designed with two proximity sensors to detect when the ceramic ball is in play and disable magnetic features like the Spiral Helper. The sensors are placed in locations such that they may only be triggered when the larger power ball is dispensed, thus marking the ball in play [16]. Since magnets are core part of detecting pinballs on most tables, ceramic balls are rendered useless on machines not specifically designed to observe nonmagnetic balls.

Another machine that famously does not use standard pinballs is *Viper Night Drivin* by Sega. *Viper Night Drivin* features a ball called the “glo-ball” which is composed of yellow, glow-in-the-dark rubber. The glo-balls are bouncier and weight approximately 25% less than standard pinballs [8]. While the idea seemed novel at the time, many players reported frustration with the bounciness of the balls making the game difficult to play. In addition, the rubber coating would get covered in dirt very quickly, and the glow-in-the-dark effect could fade after only weeks of play [6]. Another example of a nonstandard ball comes from *Bally's Truck Stop*. A ramp was accidentally included that was too narrow for normal, 1 1/16th inch pinballs. Rather than redesigning and installing new ramps on the existing machines, the game was sold with 1-inch pinballs that worked with existing ramps instead [4].



The unique “glo-ball” from *Viper Night Drivin*. [13]

While the average player will never think about Rockwell hardness or diametric deviation, the standards of the 1 1/16" pinball provide the predictable physics and interoperability our brains subtly require for high-level play. By settling on a universal size and weight, the industry has ensured that a ball manufactured in one decade can fly through a ramp designed in another. Beyond the arcade, these invisible standards are what allow a globalized world to function. As seen with the glo-ball or power ball, breaking these standards can lead to frustration. While innovation is vital, the foundation of any great game lies in consistency, and even subtle changes may cause disruption. Small differences like chrome steel or carbon steel give players control, while major variations like ceramic balls can lead to confusion and frustration. Standards in both pinball and real life should not be seen as a creative limitation, but rather a time-tested stage from which the innovations of tomorrow can be built upon.

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